

Disagreement, Agreement, and Elaboration in Crowdsourced Deliberation: Ideation Through Elaborated Perspectives

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ABSTRACT

In this study, we examined disagreement, agreement, and elaboration (rationale sharing) and their association with idea generation in a crowdsourced deliberation that took place within a crowdsourced policymaking process led by a national government. We analyzed the comments posted to the crowdsourced deliberation process and found that the elaboration of perspectives was a key element in idea generation. Disagreement contributed to ideation most when it was elaborated—i.e., when the participants justified their stances—and when it was accompanied by elaborated agreement. The findings suggest that in the design of the technologies and processes facilitating crowdsourced policymaking and other applications for civic engagement, there should be a particular focus on encouraging elaboration because elaboration can contribute to productive ideation as well as constructive argumentation. Elaboration could be fostered by deploying features from deliberation and argumentation technologies, which are designed to encourage participants to elaborate on their stances.

CCS CONCEPTS

• **Human-centered computing** → *Collaborative and social computing systems and tools*; **Computer supported cooperative work**; *Collaborative content creation*; **Empirical studies in collaborative and social computing**; Empirical studies in HCI.

KEYWORDS

civic technology, crowdsourced policymaking, deliberation, policymaking, civic participation, civic engagement, participatory democracy, democratic innovations

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1 INTRODUCTION

Policy discussions increasingly occur online in digital debates, virtual town halls, and other participatory platforms [28, 30, 42, 44]. Crowdsourced policymaking is one digital application of participatory democracy in which local and national governments engage citizens and gather ideas and knowledge to improve policies [5, 50]. Crowdsourced policymaking provides spaces for crowdsourced deliberation, which is an asynchronous, distributed, and self-selected form of deliberation [3, 36]. Crowdsourced deliberation seeks to enhance policy deliberations by uniting key elements from democratic deliberation (reasoned argument exchange characterized by respect and reciprocity among the participants) and idea crowdsourcing (gathering idea contributions from online crowds). Crowdsourced deliberation thus has a dual goal: fostering good quality deliberation and producing ideas to be considered for the policy.

In this paper, we focus on three integral discursive features in policy deliberations: disagreement, agreement, and elaboration. On the one hand, constructive disagreement can make deliberation productive by contributing to the development of reasoned perspectives, new ideas, and compromises [21, 22, 27, 46]. Disagreement has also been found to contribute to ideation in crowdsourced ideation processes [43]. On the other hand, disagreement can also lead to opinion polarization in deliberation [49, 54]. In addition to disagreement, agreement is also an important element in deliberation as an ideal outcome (i.e., group consensus) [22, 54, 56]. Discussants must reach agreement to some extent to develop compromises, and agreement helps to moderate disagreement so that it remains civil and productive [40, 56]. Elaboration, or rationale sharing, is the third essential element in constructive deliberation [21, 54, 55] because it provides discussants with arguments by which they can assess the relative merits and drawbacks of policy proposals. Elaboration may also be conducive to more productive ideation in collaborative innovation processes [48, 62].

While some studies have examined discursive processes in policy deliberations [54], such processes remain understudied, especially in a crowdsourced policymaking context. In addition to fostering deliberative quality through constructive argumentation, crowdsourced deliberation also serves to gather a large number of ideas from participants. Therefore, it is important to examine factors influencing idea production in crowdsourced deliberation.

In this paper, we pose the following *research question*: **What are the roles of disagreement, agreement, and elaboration in idea generation in crowdsourced deliberation?** To address this question, we used data from a crowdsourced law reform process led by the government of Finland. We developed a coding scheme

to analyze and categorize the crowdsourced comments and then analyzed the comment data with inferential statistics.

This paper contributes to our understanding of crowdsourced deliberation as an increasingly common form of deliberation and can help to inform the design of processes and tools that facilitate crowdsourced policymaking. This in turn can help to design and execute more productive crowdsourced policymaking processes, which are more likely to achieve the dual goals of crowdsourced deliberation: productive ideation and constructive argumentation of high deliberative quality. Furthermore, in the CSCW community, as well as in the broader HCI community, there is a growing body of work on civic engagement in applications of participatory democracy, such as civic technologies [8, 15, 52, 57, 59], deliberation and argumentation technologies [13, 29, 31, 45], and crowdsourcing for civic purposes [38, 39]. These processes and technologies are often designed to foster both ideation and deliberation with constructive argumentation. This study provides valuable knowledge to both CSCW and HCI communities about the factors affecting ideation and deliberation, thus contributing to more effective designs for processes and technologies facilitating crowdsourced policymaking and other applications of participatory democracy.

2 KEY CONCEPTS AND RELATED WORK

2.1 Crowdsourced Policymaking and Crowdsourced Deliberation

In crowdsourced policymaking, national and local governments invite citizens to participate in a policymaking process by submitting their ideas online. Governments use crowdsourcing to achieve two primary goals: (1) to develop better policies by leveraging citizens' knowledge and insights and (2) to promote civic engagement [1]. However, even when crowdsourcing is deployed in a policy reform process, elected representatives, such as members of parliament, still decide the goals of the crowdsourcing process and how to use its outcomes. Crowdsourcing is therefore a method for participatory democracy—in which the goal is to engage citizens in political processes between elections [47]—rather than for direct democracy [18] because the participants do not have decision-making power over the policy. On the ladder of citizen participation [6], crowdsourced policymaking falls on the "consultation" level, several steps removed from full citizen control. However, crowdsourcing can help to incorporate non-experts, local knowledge, and new perspectives into the policymaking process and arguably has the potential to make policymaking more inclusive, transparent, and collaborative [2, 3, 35, 42, 50].

Crowdsourced policymaking provides spaces for crowdsourced deliberation, a type of online deliberation [3] that is distributed, asynchronous, and depersonalized and therefore takes place at the "deliberative systems" level [46]. A government initiates and oversees a crowdsourced deliberation as one component of a larger policy reform process, which distinguishes crowdsourced deliberation from online deliberation in other fora, such as newspapers' comments sections or Reddit [4]. Crowdsourced deliberation has been theorized to generate new ideas, refine existing or emerging ideas, and yield better solutions to policy issues by recombining aspects of different proposals [3, 36, 37]. Crowdsourced deliberation derives from traditional forms of democratic deliberation,

which entails the fair and equal consideration of well-reasoned arguments [14, 22], characterized by mutual respect, reciprocity, and civility [21]. Unlike traditional forms of deliberation, crowdsourced deliberation leverages online platforms to accommodate a much larger number of participants, thus potentially enabling policymakers to gather a larger and more diverse set of ideas to improve policy. The increased scale of participation at low cost and amplified potential to gather knowledge to improve policy can make crowdsourced deliberation an appealing civic engagement method for governments.

Self-selection is one key factor differentiating crowdsourced policymaking from several other participatory democracy methods involving deliberation, such as deliberative polls [17] and citizens' assemblies, in which participants are selected by organizers via random or quota sampling. Because participants in crowdsourced deliberation are self-selected, they are unlikely to constitute a representative sample of the opinion of a larger population, so the results cannot serve as a proxy for "public opinion." The distinctive features of crowdsourced deliberation affect the processing and usage of the crowd's input. Typically, policymakers synthesize the crowd's contributions into a summary, which serves as one data point among many that policymakers consider [3].

Crowdsourced deliberation takes place on online platforms that are typically designed either for idea generation, on the one hand, or deliberation and argumentation, on the other, rather than integrating features from both [1–3, 26]. Ideation platforms such as IdeaScale, Spigit, and Open Town Hall [58] are designed primarily for gathering a large number of ideas rather than for soliciting justifications for ideas. In contrast, deliberation and argumentation tools, such as Consider.it and Reflect [33, 34], are designed to facilitate deliberation and consensus building by prompting participants to provide rationales when expressing their opinions or offering proposals, thus supporting the goals of traditional democratic deliberation initiatives [32, 51].

2.2 Disagreement, Agreement, and Elaboration in Deliberation

In deliberation, disagreement occurs when discussants express a perceived difference of opinion, values, or beliefs [54]. By eliciting rationales and evidence, disagreement can contribute to participants' ability to generate reasons for their opinions and the development of mutual understanding among discussants [21, 40, 49]. Disagreement has also been found to contribute to new idea generation in crowdsourced, large-scale ideation processes [43]. Disagreement is more likely to be beneficial when it is expressed in a civil manner and accompanied by elaboration, i.e., when the discussants share their rationales for their arguments, ideas, and claims [7, 21, 40]. Both disagreement and agreement play important roles in deliberation [54, 56]. Expressions of agreement help discussants establish common ground, mutual respect, and a willingness to compromise in policy deliberations and in group problem-solving [40, 54, 56]. Expressions of agreement that accompany expressions of disagreement can mitigate the potential for disagreement to be perceived as impolite [40, 54, 56], thus helping to maintain civility and mutual respect despite differences of opinion.

Elaboration, or rationale sharing, is an essential element of both disagreement and agreement in deliberative processes because it provides discussants with material by which they can judge the merits of and potential justifications for a policy proposal. An elaboration is a statement with additional argumentation or information. Elaboration can be, for example, in the form of further justification, a definition, or an example [54]. Elaboration has also been identified as important in ideation processes in small group collaboration because elaboration helps build common ground and a shared mental model [48, 62]. In a similar vein, research on ideation and innovation processes has shown that awareness of other discussants' reasoning during ideation helps group members generate more ideas [60, 61]. Elaboration thus helps deliberation fulfill its democratic potential and can also simultaneously support idea generation, another key goal of crowdsourced deliberation. In idea crowdsourcing, participants' exposure to each others' rationales can help participants to establish common ground, raise awareness of each other's knowledge, and improve the average quality of ideas [61, 62]. Thus, elaboration can support deliberative democratic ideals of communicative inclusion and reasoned argumentation as well as idea generation.

Taken together, disagreement, agreement, and elaboration are central elements in deliberation [21, 54] and have also been identified as factors potentially contributing to ideation in innovation processes [26, 48, 62]. These discursive features of deliberation can contribute to new idea generation, improve the quality of generated ideas, enhance understanding of problems, and help to refine ideas by prompting rational justification [26, 43, 62]. However, the roles of disagreement, agreement, and elaboration in ideation in crowdsourced deliberation remain understudied. Knowing more about the roles of disagreement and elaboration in idea generation can help us better understand the dynamics of crowdsourced deliberation and improve the design of processes and technologies for more productive crowdsourced deliberation. To examine the relationship between disagreement, elaboration, and idea generation in crowdsourced deliberation, we posed the following **research question: What are the roles of disagreement, agreement, and elaboration in idea generation in crowdsourced deliberation?** To address this question, we first developed a coding scheme to analyze comment data from a crowdsourced deliberation led by the Ministry of Justice in the government of Finland. We then further analyzed the data using a multivariate linear regression model and a multi-step inferential analysis, which we describe in more detail in the next section.

3 CASE PROFILE, METHODS, AND DATA

3.1 Crowdsourced Association Act Reform

3.1.1 Case Profile. Finland's Ministry of Justice launched the crowdsourced policymaking process this study examines in May 2019. The Ministry sought to crowdsource ideas for reforming a law called the Association Act (26.5.1989/503), which governs established forms of civic engagement, namely, the establishment and management of associations. The crowdsourced deliberation took place on a crowdsourcing platform called HowSpace, which was open for anyone to participate. The participants could access by signing up with an email address and could choose to remain anonymous or use their

real names on the platform. Three representatives of the Ministry of Justice participated in the deliberation and post-moderated the interactions. The representatives stated that they did not remove any comments in moderation. The Ministry announced the crowdsourcing initiative through various digital channels, including a website dedicated to the Association Act reform and social media. The Ministry formulated three main topic areas for deliberation on the crowdsourcing platform. Participants were prompted to either post their own ideas for improving the law or react to proposals developed by the Ministry representatives. The participants could post their ideas, reply to existing posts, and like others' proposals or comments. The platform represented a large-scale idea crowdsourcing technology rather than a deliberation technology because there were no specific features for mapping arguments or synthesizing pros and cons of proposals. The crowdsourced deliberation lasted four weeks in June 2019. 151 participants took part, and 488 comments were submitted on the platform. When the crowdsourcing process had concluded, the Ministry representatives analyzed the crowd's input and published the results in a report. The reformed law came into effect in February 2023.

3.2 Methods and Data

3.2.1 Crowdsourced Comments and Coding Scheme. The comment data (488 comments) were exported from the platform and translated into English. To analyze the data, we developed a coding scheme based on studies of crowdsourcing [2, 3], deliberation [17, 37], and disagreement [16, 54, 56]. The categories in the coding scheme are described in Table 1 in the appendix, with definitions and examples for each code. We applied the initial coding scheme to a sample of 150 comments, which represented 30% of the total number of comments. Three researchers first coded the pilot data individually. We then discussed any discrepancies. To resolve discrepancies, we consulted the literature, discussed the best approach to address our research question, and refined the coding scheme accordingly. We repeated this revision process until we reached consensus on the final version of the scheme. Two researchers then coded each comment for the presence or absence of *disagreement*, *agreement*, *elaboration*, and at least one *new idea*. We also counted the *number of ideas* presented in each comment. Table 1 in the appendix provides a definition and examples for each coding category. We used Krippendorff's alpha to check intercoder reliability between the two coders. We report the *k.alpha* for each code in Table 1. Table 2 in the appendix shows the frequencies of occurrence of disagreement, agreement, elaboration and new ideas in the data. In the comment data analysis, we included posts by the discussion facilitators (three representatives of Finland's Ministry of Justice), which constituted about 25% of the total number of comments. We decided to do so because of the role of the Ministry representatives in this crowdsourced deliberation: They contributed to the deliberation in a manner similar to the participants rather than as supervisors.

3.2.2 Multivariate linear regression model for online comment data analysis. To examine the relationship between disagreement, elaboration, and idea generation, we developed a multivariate linear regression model, which is presented in Table 3 in the appendix. In this model, a comment was the unit of analysis, and altogether,

488 comments were analyzed. Elaboration was operationalized as a type of dis/agreement: each comment featuring agreement or disagreement was categorized as either simple or elaborated according to the coding scheme (see Section 3.2.1 and Table 1 in the appendix). We therefore included the independent variables *simple agreement*, *simple disagreement*, *elaborated agreement*, and *elaborated disagreement* in the regression model. The model used *number of ideas* as the dependent variable. We checked for multicollinearity among the independent variables to ensure that correlated variables were not included in the model. Multicollinearity complicates the correct estimation of the relationship between each independent variable and the dependent variable [9]. Similar to [10, 20, 53], variance inflation factors (VIFs) were computed to detect multicollinearity among the independent variables. All VIFs were below 5, indicating the nonexistence of linear dependency among the independent variables [24].

3.2.3 Multi-step inferential analysis: Disagreement, agreement, elaboration, and idea generation. To examine how the type of dis/agreement (simple or elaborated) and the presence/absence of agreement and disagreement are associated with the number of ideas in crowd-sourced deliberation, we complemented the regression analysis with a multi-step inferential analysis. We compared the numbers of ideas in the comments, which were grouped based on the presence of disagreement and/or agreement and the type of dis/agreement (simple or elaborated). The design of this procedure is illustrated in Figure 1 in the appendix. In *Step 1*, we divided the 488 comments into two sets: one containing comments in which disagreement and/or agreement occurred and the other containing comments featuring neither agreement nor disagreement. We computed the mean number of ideas in each group. Because the number of ideas in the groups of comments did not follow a normal distribution, we used the non-parametric Mann–Whitney U test [41] to calculate the statistical significance of the difference in the means. *Step 1* enabled us to check (1) whether comments featuring disagreement and/or agreement had on average more ideas than comments without agreement or disagreement and (2) whether the differences were statistically significant. However, this step did not provide enough granularity to distinguish whether a potential difference in the number of ideas came from comments featuring agreement, comments featuring disagreement, or comments in which both agreement and disagreement were present. Therefore, in *Step 2*, comments featuring disagreement and/or agreement were subdivided into three sets: (1) comments featuring only agreement, (2) comments featuring only disagreement, and (3) comments featuring both agreement and disagreement. As in *Step 1*, we calculated the mean number of ideas in each group. We used the non-parametric Kruskal-Wallis one-way ANOVA test [11] to check whether the differences in the means of the groups of comments were statistically significant. Tukey’s post hoc test was used to determine which differences between the groups’ means were statistically significant (see *Step 2* in Figure 1 in the appendix).

Based on the results from *Step 2*, we were able to verify an association between the presence of dis/agreement and the number of ideas. However, we were not able to examine the role of the type of dis/agreement (simple or elaborated). Therefore, we took

a third and final step to examine the roles of simple and elaborated dis/agreement in idea generation. In *Step 3*, we divided the comments featuring only agreement into two subsets: one comprising comments that featured only simple agreement and another comprising comments that featured only elaborated agreement. Similarly, the comments featuring only disagreement were separated into two subsets: one comprising comments that featured only simple disagreement and the other comprising comments that featured only elaborated disagreement. The comments in which agreement and disagreement co-occurred was divided into four subsets, each containing one possible combination of simple/elaborated dis/agreement: i) simple agreement and simple disagreement, ii) simple agreement and elaborated disagreement, iii) elaborated agreement and simple disagreement, and iv) elaborated agreement and elaborated disagreement. As in *Step 2*, the mean number of ideas in each group was computed, and the Kruskal-Wallis one-way ANOVA and Tukey’s post hoc tests were used to determine whether the differences between the means were statistically significant (see *Step 3* in Figure 1).

4 RESULTS

4.1 Multivariate Regression Analysis

The multivariate regression model with the comment analysis codes *simple agreement*, *elaborated agreement*, *simple disagreement*, and *elaborated disagreement* as variables, as described in Section 3.2.2, is presented in Table 3 in the appendix. The results show that the presence of new ideas was positively and statistically significantly associated with *elaborated agreement* and *elaborated disagreement*. Discussions in which elaborated agreement or elaborated disagreement was present tended to produce 0.5 more new ideas than comments that did not feature elaborated agreement or elaborated disagreement. Simple agreement had a statistically significant but negative association with the presence of ideas in the comments. This factor was negatively correlated with the dependent variable *new ideas* with a magnitude of -0.19, as shown in Table 3 in the appendix.

4.2 Multi-Step Inferential Analysis

To analyze the association between simple and elaborated dis/agreement and idea generation by applying multi-step inferential analysis, we compared comment groups through a three-step process, as described in Section 3.2.2. We present the results in the following. As shown in Table 4 in the appendix, 410 out of the 488 comments (84%) featured agreement and/or disagreement. These comments produced on average 1.56 ideas. On the other hand, comments featuring neither agreement nor disagreement ($n=78$) generated, on average, 1.17 ideas. *Step 1* in the analysis showed that the difference between these means was statistically significant ($p\text{-value} < 0.001$) based on the Mann–Whitney U test. This result indicates that comments containing agreement and/or disagreement featured, on average, a statistically significantly greater number of ideas than comments without disagreement or agreement. In *Step 2*, we measured the differences between the mean numbers of ideas in the groups of comments with only agreement, only disagreement, and both agreement and disagreement (see comment groups 3, 4, and 5 in Table 4 in the appendix). The Kruskal-Wallis one-way ANOVA

showed that the differences between the means of the comment groups were statistically significant (p -value < 0.001). A Tukey's post hoc analysis of these groups showed that comments that featured both disagreement and agreement contained a statistically significantly greater number of ideas than comments with only agreement (p -value < 0.01) and comments with only disagreement (p -value < 0.05). In Step 3, we tested the differences between the mean numbers of ideas in comments featuring only simple agreement, only elaborated agreement, only simple disagreement, only elaborated disagreement, and both elaborated agreement and elaborated disagreement (see comment groups 6, 7, 8, 9, and 10 in Table 4). As outlined in Table 4, the comment groups featuring simple agreement and simple disagreement ($n=11$), simple agreement and elaborated disagreement ($n=12$), and elaborated agreement and simple disagreement ($n=13$) were not large enough to be included in the analysis. The Kruskal-Wallis one-way ANOVA indicated that the remaining groups of comments (6, 7, 8, 9, and 10) differed statistically significantly in the average numbers of ideas they featured (p -value < 0.001). A Tukey's post hoc analysis showed that comments featuring both elaborated agreement and elaborated disagreement produced on average a statistically significantly greater number of ideas than comments featuring simple disagreement and simple agreement (p -value < 0.01, p -value < 0.01, respectively) and comments featuring only elaborated disagreement (p -value < 0.01). Tukey's test also showed that comments featuring elaborated agreement had a statistically significantly greater number of ideas than comments featuring simple disagreement and simple agreement (p -value < 0.01, p -value < 0.01, respectively). Similarly, the post hoc test indicated that comments featuring elaborated disagreement contained a statistically significantly greater number of ideas than comments featuring simple disagreement and agreement (p -value < 0.01, p -value < 0.01, respectively). These results indicate that 1) comments featuring disagreement and agreement contained more ideas than comments without disagreement or agreement; 2) comments featuring elaborated disagreement or elaborated agreement had more ideas than comments with only simple disagreement or simple agreement; and 3) comments featuring *both* elaborated disagreement and elaborated agreement contained more ideas than comments featuring only simple disagreement or simple agreement.

5 LIMITATIONS

The main limitation of this study is that it is based on one crowdsourced policymaking process. Having several crowdsourced policymaking processes in the sample would diversify and expand the scope of the policy issues under consideration, which could affect the discursive dynamics in crowdsourced deliberation. However, studying several crowdsourced policymaking processes simultaneously is not a trivial challenge from a data gathering perspective. Furthermore, many empirical studies about crowdsourced policymaking are based on one case [3, 12, 23, 25, 30], which has been identified as a valuable approach in recent scholarship on crowdsourcing [19]. Moreover, despite the limited scope of a single crowdsourced policymaking process in this study, we believe that the results make a valuable contribution to the body of knowledge about crowdsourced deliberation and other processes that include

similar elements, such as crowd participation, idea generation, and deliberation.

6 DISCUSSION AND CONCLUSION

We examined the roles of disagreement, agreement, and elaboration in ideation in crowdsourced deliberation. We found that the simultaneous presence of disagreement and agreement in comments was more strongly positively associated with idea generation than the presence of either disagreement or agreement alone or the absence of both. Moreover, the simultaneous presence of elaborated disagreement and elaborated agreement had the strongest positive association with idea generation. This suggests that elaboration accompanying dis/agreement—already considered an important marker of deliberative quality [21, 22, 54] and a factor that supports ideation in collaborative innovation processes [43, 62]—can also contribute to idea generation in crowdsourced deliberation. By providing shared ground and material for idea generation, elaboration associated with dis/agreement in crowdsourced deliberation can play a similar role as in more traditional forms of deliberation and collaborative innovation processes [26, 48, 61].

The findings also show that disagreement is not necessarily a negative factor in crowdsourced deliberation—disagreement can actually play a constructive role as long as it is expressed in a civil manner and elaborated, similar to the role of disagreement in more traditional deliberation settings [7, 21, 40]. Crowdsourced policymaking processes therefore should not discourage disagreement but rather harness its productive power by encouraging participants to elaborate when they express disagreement. Moreover, the findings suggest that comments that express both agreement and disagreement may contribute to ideation by signaling respect for others' opinions, thus encouraging them to share their ideas (or at least not discouraging idea expression), and because such comments are more likely to feature justifications or rationales [40]. These results suggest implications for designing technologies for civic engagement, whether for crowdsourced policymaking or for other participatory democracy practices with the goals of ideation and constructive argumentation.

The findings suggest prioritizing elaboration for both productive ideation and constructive argumentation. This could be done by deploying features from deliberation and argumentation technologies [2, 32, 33], which are designed to encourage participants to elaborate on their stances. Making key points of agreement and disagreement in a crowdsourced deliberation more salient and easily accessible to participants could help to stimulate ideation. Platform interface design interventions and facilitator interventions could help to achieve this.

This study is one of the first steps towards improving our understanding of crowdsourced deliberation, especially the roles of disagreement, agreement, and elaboration in crowdsourced deliberation and deliberative processes at large. Future research should further examine the relationship between the discursive features that occur in deliberation and idea generation. It would be particularly valuable to examine the role of elaboration in idea generation further by developing a more nuanced approach to the types of elaboration.

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A APPENDIX: TABLES AND FIGURES

Table 1: Description of the coding scheme: Code categories, values, definitions, and examples

Name of code	Definition	Example
<i>Disagreement, Agreement, Elaboration</i>		
Disagreement (values: 0/1; Inter-coder reliability (IR)= 0.89)	When the participant expressed disagreement with a proposal or idea	"I do not support the proposal." (simple disagreement)
Agreement (values: 0/1; IR:=0.89)	When a participant signaled support for another participant's or facilitator's comment	"I support the proposal." (simple agreement)
Elaboration (values: 0/1; IR=0.95)	Any expression of disagreement or agreement was coded as either simple or elaborated; an expression of dis/agreement was categorized as simple when it expressed dis/agreement without any elaboration and as elaborated according to the definition in [54] (see Section 2.2)	"I think it's a bad solution, because nobody remembers this sort of distinction in the daily practice of association activities." (elaborated disagreement). "I support this. Openness is always a good thing, and when there's a provision on it, members have something they can invoke." (elaborated agreement)
<i>Idea Generation</i>		
New idea (values: 0/1; IR=0.95)	Comment contributed at least one new idea to the discussion	"I think it would be a good idea to allow online meetings, and mention that in the law for clarity's sake, but leave the actual issue to the rules of the association. Then those associations that believe that their members would have an equal opportunity to participate could change their rules."
Number of ideas (values: numeric)	The number of distinct ideas was counted for each comment	"There should be several founders so that there would be more than one person monitoring the activities, already because of potential abuses." (number of ideas=1)

Table 2: Frequencies of the types of disagreement, agreement, and elaboration in online comments.

Coding category	Number of comments	Percentage of total	Percentage within category*
Disagreement	278	56.97	
– Simple disagreement	26	5.33	9.35
– Elaborated disagreement	252	51.64	90.65
Agreement	246	50.41	
– Simple agreement	54	11.07	21.95
– Elaborated agreement	192	39.34	78.05

Note. The percentages for “disagreement” and “agreement” do not sum to 100 because both of these codes could be applied to a single comment. * The codes “simple” and “elaborated,” which are mutually exclusive, could only be applied to comments featuring disagreement and/or agreement, so these are presented as percentages of their respective category (e.g., elaborated disagreement is given as a percentage of comments featuring disagreement)

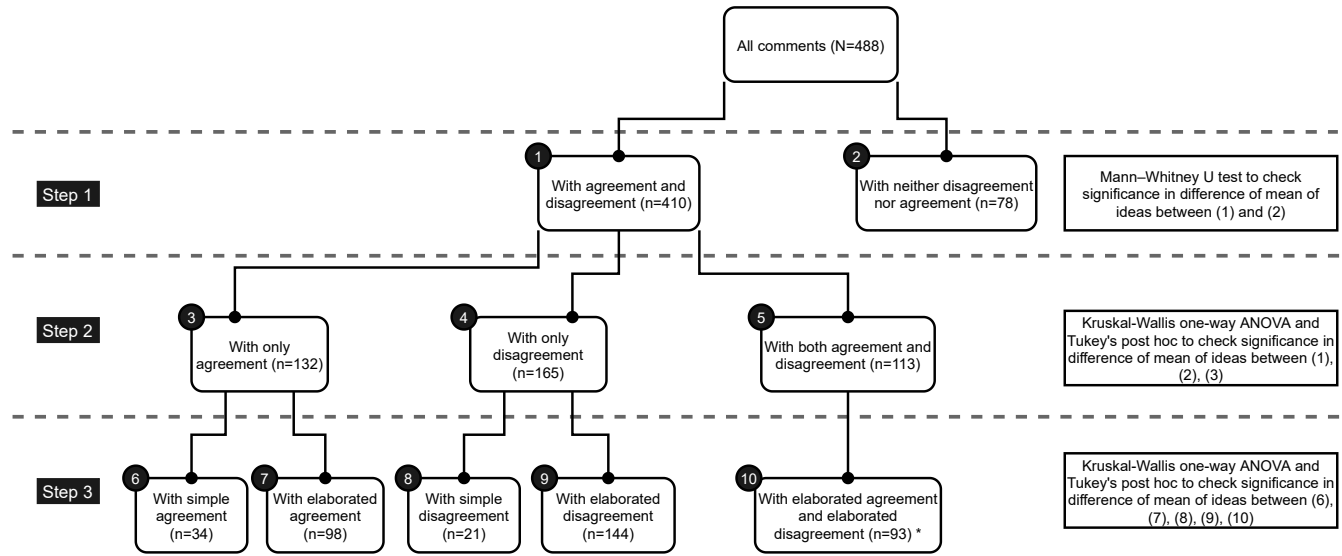
Table 3: Multivariate linear regression model for analyzing the association between new idea generation and the types of dis/agreement.

Variable	Coefficient	Std err	t	p-value
(Intercept)	1.13	0.08	13.58	0.000***
Simple agreement	-0.51	0.14	-3.63	0.000**
Elaborated agreement	0.50	0.10	5.50	0.000**
Simple disagreement	-0.19	0.20	-0.96	0.34
Elaborated disagreement	0.50	0.10	5.63	0.000**

** p-value <0.01 *** p-value <0.001, $N = 488$, $R^2 = 0.18$

Table 4: Numbers of comments and means of the number of ideas in online comments grouped according to the type(s) of dis/agreement (simple/elaborated) they featured.

	Comment group	n	Mean of ideas
Step 1			
1	Agreement and/or disagreement	410	1.56
2	Neither agreement nor disagreement	78	1.17
Step 2			
3	Only agreement	132	1.36
4	Only disagreement	165	1.52
5	Both agreement and disagreement	113	1.97
Step 3			
6	Only simple agreement	34	0.35
7	Only elaborated agreement	98	1.70
8	Only simple disagreement	21	0.76
9	Only elaborated disagreement	144	1.63
10	Elaborated agreement and elaborated disagreement	93	2.07
11	Simple agreement and simple disagreement	5	1.20
12	Simple agreement and elaborated disagreement	15	1.47
13	Elaborated agreement and simple disagreement	0	—



* Among comments featuring both disagreement and agreement, only the subgroup of comments featuring both elaborated disagreement and elaborated agreement was large enough to analyze in comparison with other subgroups. The other subgroups (simple agreement and simple disagreement; simple agreement and elaborated disagreement; and elaborated agreement and simple disagreement) each featured 15 comments or fewer and were therefore not included in comparative analyses

Figure 1: Design of the multi-step inferential analysis (see Section 3.2.3) to examine the roles of disagreement and agreement in idea generation in crowdsourced deliberation.